

DSA + LLD Daily & Weekly Planner

Atin Sharma · SDE-1 · 8-Week Arc · Weekends light, buffers built in

How to use this planner

Three layers. The **Weekly Master Planner** gives every day a specific assignment, which is your daily task list. The **Daily Time-Block Templates** tell you *when* to do each thing (weekday, Saturday, Sunday rhythms differ). The **[Added] Concepts** section covers five gaps I found in the original sheet, each tagged **[Added]** so you can tell them apart. Read the **Flexibility Rules** at the end before you start, they are what keep this from being abandoned.

What I added to the original DSA roadmap

Concept	Tag	Priority	Why it earns a slot
Maximum Subarray (Kadane's, LC53)	[Added]	Must	Foundational, was missing entirely
Union-Find / DSU	[Added]	Must	Graph problems need it; never taught standalone
Tries (LC208/211)	[Added]	Must	Amazon autocomplete-style questions
Bit Manipulation	[Added]	Light	Occasional Amazon/fintech; weekend filler
Matrix Simulation	[Added]	High	LC48/LC54 are Amazon SDE-1 favorites

Weekly Master Planner — Each Day's Specific Assignment

Weekday DSA 6–8:30 AM with Srishti · LLD evening 9–10 PM · Sat/Sun shaded = light

Week 1 Trees finish + Two Pointers entry · LLD: SOLID principles

Day	DSA — Morning (6:00–8:30 AM)	LLD — Evening (9–10 PM)	Platform
Mon	Trees: LC230 Kth Smallest BST, LC235 LCA of BST. Concept: inorder of BST = sorted.	SOLID: SRP + OCP. Write 1 violation + 1 fix each in TS.	TUF+ A2Z (Trees) → LC TUF+ LLD L3-4
Tue	Trees: LC98 Validate BST, LC105 Construct from Pre+Inorder.	SOLID: LSP + ISP.	TUF+ A2Z → LC TUF+ LLD L5-6
Wed	Two Pointers: LC125 Valid Palindrome, LC167 Two Sum II.	SOLID: DIP + recap all 5.	TUF+ A2Z (2-Ptr) → LC TUF+ LLD L7
Thu	Two Pointers: LC15 3Sum, LC11 Container With Most Water.	UML: class diagram basics. Redraw the example.	TUF+ A2Z → LC TUF+ LLD L8
Fri	Two Pointers: LC42 Trapping Rain Water (hard — 1 problem) + buffer.	UML: draw a sample system from scratch on Excalidraw.	TUF+ A2Z → LC TUF+ LLD L9
Sat	LIGHT — Re-do ToDoAgain: LC49 Group Anagrams, LC347 Top K, LC128 Longest Consecutive.	Off.	LC (re-do, no timer)
Sun	BUFFER — finish anything that slipped this week. Scan tracker, plan Week 2.	Off.	—

Week 2 Sliding Window + Binary Search · LLD: Creational patterns

Day	DSA — Morning (6:00–8:30 AM)	LLD — Evening (9–10 PM)	Platform
Mon	Sliding Window: finish LC3 Longest Substring + LC567 Permutation in String (both in-progress).	Intro to Design Patterns + Singleton (thread-safe variant).	TUF+ A2Z (SW) → LC TUF+ LLD L10-11
Tue	Sliding Window: LC76 Min Window Substring (hard), LC424 Longest Repeating Char Replacement.	Factory Method (use a string-keyed registry).	TUF+ A2Z → LC TUF+ LLD L12
Wed	Binary Search: LC704 Classic (template anchor), LC33 Search in Rotated Sorted Array.	Builder Pattern (fluent API in TS).	TUF+ A2Z (BS) → LC TUF+ LLD L13
Thu	Binary Search: LC153 Find Min Rotated, LC74 Search 2D Matrix [High].	Abstract Factory.	TUF+ A2Z → LC TUF+ LLD L14
Fri	Sliding Window: LC239 Sliding Window Max [High, deque] + buffer.	Recap creational. Code one pattern cold from memory.	TUF+ A2Z → LC TUF+ LLD
Sat	LIGHT — Re-do any SW/BS problem scored under 3 this week.	Off.	LC (re-do)
Sun	BUFFER — review + plan Week 3.	Off.	—

Week 3 Linked List + Stack · LLD: Structural patterns

Day	DSA — Morning (6:00–8:30 AM)	LLD — Evening (9–10 PM)	Platform
Mon	Linked List: LC21 Merge Two Sorted, LC141 Cycle, LC876 Middle. Master fast/slow pointers.	Adapter Pattern.	TUF+ A2Z (LL) → LC TUF+ LLD L16
Tue	Linked List: LC142 Cycle II (entry point), LC19 Remove Nth From End.	Decorator: chain Logger + Cache around a service.	TUF+ A2Z → LC TUF+ LLD L17
Wed	Linked List: LC143 Reorder List, LC146 LRU Cache [High — HashMap + DLL].	Facade Pattern.	TUF+ A2Z → LC TUF+ LLD L18
Thu	Stack: LC20 Valid Parentheses, LC155 Min Stack, LC496 Next Greater Element.	Proxy Pattern.	TUF+ A2Z (Stack) → LC TUF+ LLD L20
Fri	Stack: LC739 Daily Temperatures (mono stack), LC84 Largest Rectangle (hard).	Recap structural. Skip Composite/Bridge/Flyweight.	TUF+ A2Z → LC TUF+ LLD
Sat	LIGHT — [Added] Union-Find: learn DSU template (find + union by rank), then LC323 Connected Components, LC547 Number of Provinces.	Off.	TUF+ A2Z (Graph DSU) → LC
Sun	BUFFER — review + plan Week 4.	Off.	—

Week 4 Graphs + Heap · LLD: Behavioural patterns

Day	DSA — Morning (6:00–8:30 AM)	LLD — Evening (9–10 PM)	Platform
Mon	Graphs: LC200 Number of Islands, LC695 Max Area of Island (DFS grid template).	Observer: build EventEmitter from scratch.	TUF+ A2Z (Graph) → LC TUF+ LLD L24
Tue	Graphs: LC994 Rotting Oranges (multi-source BFS), LC133 Clone Graph.	Strategy: payment processor swapping UPI/Card/Wallet.	TUF+ A2Z → LC TUF+ LLD L25
Wed	Graphs: LC207 Course Schedule, LC210 Course Schedule II (Kahn's topo sort).	Command: Undo/Redo with command stack.	TUF+ A2Z → LC TUF+ LLD L26
Thu	Graphs: LC417 Pacific Atlantic [High], LC684 Redundant Connection [High — uses Sat's DSU].	State: TrafficLight with 3 states.	TUF+ A2Z → LC TUF+ LLD L27
Fri	Heap: LC215 Kth Largest (min-heap size K), LC973 K Closest Points.	Iterator + recap behavioural.	TUF+ A2Z (Heap) → LC TUF+ LLD
Sat	LIGHT — Heap: LC621 Task Scheduler. Then [Added] Bit Manipulation: LC136 Single Number, LC191 Number of 1 Bits.	Off.	TUF+ A2Z (Bit Manip) → LC
Sun	BUFFER — review + plan Week 5.	Off.	—

Weekly Master Planner (continued)

Week 5 Heap finish + Backtracking + Tries · LLD: Advanced reads + Parking Lot

Day	DSA — Morning (6:00–8:30 AM)	LLD — Evening (9–10 PM)	Platform
Mon	Heap: LC295 Find Median from Data Stream (two-heap technique).	Read: DI + Exception Handling (typed Result monad).	TUF+ A2Z → LC TUF+ LLD (Advanced)
Tue	Backtracking: LC78 Subsets (include/exclude), LC46 Permutations.	Read: Best Practices + API Design.	TUF+ A2Z (Backtrack) → LC TUF+ LLD
Wed	Backtracking: LC39 Combination Sum, LC40 Combination Sum II (skip dups).	Parking Lot — read requirements + draw class diagram (Steps 1-2).	TUF+ A2Z → LC TUF+ LLD Part-1
Thu	Backtracking: LC79 Word Search, LC22 Generate Parentheses.	Parking Lot — write full TS code (Step 3).	TUF+ A2Z → LC TUF+ LLD Part-1
Fri	[Added] Tries: LC208 Implement Trie (Must), LC211 Add and Search Word [High].	Parking Lot — compare with solution, write 3-bullet gap list (Step 4).	TUF+ A2Z (Trie) → LC TUF+ LLD
Sat	LIGHT — Re-do backtracking problems scored under 3. Then [Added] Matrix: LC54 Spiral Matrix.	Off.	LC (re-do + new)
Sun	BUFFER — review + [Added] Matrix: LC48 Rotate Image. Plan Week 6.	Off.	LC

Week 6 Greedy + DP 1D start · LLD: Vending Machine + re-do Parking Lot

Day	DSA — Morning (6:00–8:30 AM)	LLD — Evening (9–10 PM)	Platform
Mon	Greedy: LC55 Jump Game (max reach), LC45 Jump Game II (min jumps).	Vending Machine — read + diagram (Steps 1-2).	TUF+ A2Z (Greedy) → LC TUF+ LLD Part-1
Tue	Greedy: LC134 Gas Station, LC252 Meeting Rooms I, LC435 Non-overlapping Intervals.	Vending Machine — code (State pattern: Idle/HasMoney/Dispensing/OutOfStock).	TUF+ A2Z → LC TUF+ LLD
Wed	Greedy: LC56 Merge Intervals + [Added] LC53 Maximum Subarray (Kadane's — foundational Must).	Vending Machine — compare + gaps.	TUF+ A2Z → LC TUF+ LLD
Thu	DP 1D: LC70 Climbing Stairs, LC198 House Robber, LC213 House Robber II. Write recurrence BEFORE code.	Re-do Parking Lot cold from scratch (Step 5).	TUF+ A2Z (DP) → LC TUF+ LLD
Fri	DP 1D: LC322 Coin Change (unbounded knapsack), LC300 LIS.	Buffer / catch-up on LLD.	TUF+ A2Z → LC TUF+ LLD
Sat	LIGHT — Re-do greedy interval problems + recall Parking Lot diagram from memory.	Off.	LC + Excalidraw
Sun	BUFFER — review + plan Week 7.	Off.	—

Week 7 DP 1D finish + DP 2D · LLD: ATM + PubSub

Day	DSA — Morning (6:00–8:30 AM)	LLD — Evening (9–10 PM)	Platform
Mon	DP 1D: LC5 Longest Palindromic Substring (expand around center), LC91 Decode Ways.	ATM — read + diagram. Name the State pattern.	TUF+ A2Z → LC TUF+ LLD Part-2
Tue	DP 1D: LC139 Word Break, LC152 Maximum Product Subarray (track min/max).	ATM — code (State machine).	TUF+ A2Z → LC TUF+ LLD
Wed	DP 2D: LC62 Unique Paths, LC1143 Longest Common Subsequence (classic 2D template).	PubSub — read + diagram (maps to your BuilMQ work).	TUF+ A2Z → LC TUF+ LLD Part-2
Thu	DP 2D: LC72 Edit Distance (hard — insert/delete/replace).	PubSub — code the subscriber registry.	TUF+ A2Z → LC TUF+ LLD
Fri	DP: LC416 Partition Equal Subset Sum [High, 0/1 knapsack] + buffer.	Re-do Vending Machine cold.	TUF+ A2Z → LC TUF+ LLD
Sat	LIGHT — Re-do weak DP problems (lowest-scored).	Off.	LC (re-do)
Sun	BUFFER — review + plan Week 8.	Off.	—

Week 8 Mock week + cleanup · LLD: Elevator + Ride Booking

Day	DSA — Morning (6:00–8:30 AM)	LLD — Evening (9–10 PM)	Platform
Mon	MOCK — 2 timed mediums (1 weak pattern + 1 Graph/DP). No hints. Score 1-5.	Elevator — read + diagram (State + Strategy scheduling).	LC (timed) TUF+ LLD Part-3
Tue	Cleanup — clear remaining [High]: LC138 Copy List w/ Random Pointer, LC57 Insert Interval.	Elevator — code.	TUF+ A2Z → LC TUF+ LLD
Wed	MOCK — 2 timed mediums.	Ride Booking — read + diagram (anchor to VyaparGPT routing).	LC (timed) TUF+ LLD Part-3
Thu	Weak-pattern deep dive — your lowest-scored pattern, 3 problems.	Ride Booking — code.	LC TUF+ LLD
Fri	MOCK — full timed set (3-4 problems, mixed patterns).	Re-do ATM cold.	LC (timed) TUF+ LLD
Sat	LIGHT — Light review + Amazon Leadership Principles / behavioural story notes.	Off.	Notes / STAR library
Sun	BUFFER — rest + interview readiness checklist. You're ready.	Off.	—

Daily Time-Block Templates — The Rhythm of Each Day

What to open and do, hour by hour. The weekly grid says *WHAT*; this says *WHEN*.

Weekday (Mon–Fri) — full day, 2 DSA problems + 1 LLD block

Time	Block	What to do	Platform
6:00–6:10	Warm-up	Re-read yesterday's solution approach (no code). Open today's 2 problems on LeetCode in tabs.	LeetCode
6:10–6:55	DSA Block 1 (timed, with Srishti)	Problem 1 — 45-min timer. Aim to solve clean. If stuck at 30 min, peek at TUF+ approach (not code).	TUF+ A2Z → LeetCode
6:55–7:05	Break	Walk, water. Do not check phone feeds.	—
7:05–7:50	DSA Block 2 (timed)	Problem 2 — 45-min timer. Same rules.	TUF+ A2Z → LeetCode
7:50–8:00	Break	Reset.	—
8:00–8:30	Debrief (with Srishti)	Score each problem 1-5 on the rubric. Log to tracker. Mark revisit-needed. Note the ONE insight per problem.	MyTracker
9:00–10:00 PM	LLD Block	Per the weekly theme. Watch → code from memory, or run the 5-step problem drill. Skippable on heavy DP days.	TUF+ LLD

Saturday — light, one short morning block then off

Time	Block	What to do	Platform
7:00–7:45	Revision Block (no timer)	Re-do 2 problems marked revisit-needed this week. Goal: solve them clean and fast this time.	LeetCode (re-do)
7:45–8:15	[Added] concept / weak area	Light new concept per the weekly grid (Union-Find, Bit Manip, Matrix) OR practice your weakest pattern.	TUF+ A2Z → LC
Rest of day	Off	Genuinely off. Protect this — it's what keeps the plan alive past Week 3.	—

Sunday — lightest, buffer + review + plan, then rest

Time	Block	What to do	Platform
8:00–8:45	Buffer / catch-up	Finish anything that slipped during the week. This is your pressure valve — use it.	Whatever slipped
8:45–9:15	Week review	Scan tracker. Identify lowest-scored pattern. Write 1 line per weak problem on why it was hard.	MyTracker
9:15–9:30	Plan next week	Glance at next week's row in the weekly grid. Pre-open the first 2 problems in browser tabs.	This planner
Rest of day	Off	Rest fully. LLD off. No new concepts.	—

[Added] Concepts — Gaps Filled in the Original Roadmap

Five additions, each tagged. Slotted into light/weekend days so they don't add weekday load.

[Added] Maximum Subarray — Kadane's (LC53)

Priority: Must · Slots into: Week 6 Wed

Why:	Foundational Must that was absent from the original sheet. Asked everywhere, including Amazon.
Resource:	TUF+ A2Z — DP / Arrays section. Striver's Kadane explanation is the cleanest starting point.
Problems:	LC53 Maximum Subarray

[Added] Union-Find / DSU

Priority: Must · Slots into: Week 3 Sat → used Week 4 Thu

Why:	Your graph problems (Connected Components, Redundant Connection) need it, but it's never taught as a standalone template. Learn the DSU class once: find with path compression + union by rank.
Resource:	TUF+ A2Z — Graph section has dedicated DSU lessons. Start here, not Educative — Striver's template is interview-ready.
Problems:	LC323 Connected Components , LC547 Number of Provinces , LC684 Redundant Connection

[Added] Tries

Priority: Must (LC208) · Slots into: Week 5 Fri

Why:	Completely absent. LC208 Implement Trie is a genuine SDE-1 Must — autocomplete/prefix questions show up at Amazon and search-heavy startups.
Resource:	TUF+ A2Z — Trie section for the node structure, then LeetCode. Skip LC212 Word Search II (hard, low SDE-1 frequency).
Problems:	LC208 Implement Trie , LC211 Add and Search Word [High]

[Added] Bit Manipulation

Priority: Light · Slots into: Week 4-5 weekends

Why:	Absent. Amazon and fintech occasionally drop one bit-manip question. Low effort, high floor — perfect light weekend material, not worth a weekday slot.
Resource:	TUF+ A2Z — Bit Manipulation section. All easy-to-medium.
Problems:	LC136 Single Number , LC191 Number of 1 Bits , LC338 Counting Bits , LC190 Reverse Bits

[Added] Matrix Simulation

Priority: High · Slots into: Week 5-6 weekends

Why:	Absent as a category. LC48 Rotate Image and LC54 Spiral Matrix are Amazon favorites at SDE-1 — pure simulation, no deep algorithm.
Resource:	LeetCode directly — these are simulation, not pattern-heavy. TUF+ A2Z arrays-advanced covers them too.
Problems:	LC54 Spiral Matrix , LC48 Rotate Image , LC73 Set Matrix Zeroes

Flexibility & Buffer Rules — Read This Twice

1. Two problems per weekday, not four. One problem solved clean and understood beats three rushed and forgotten.
2. Sunday is your buffer valve. Miss a weekday session? Sunday absorbs it. A missed day never cascades into quitting.
3. Weekends are deliberately light — one short morning block, then off. This is not laziness; it's what keeps you going past Week 3.
4. DSA morning (6–8:30 AM) is the one fixed anchor. LLD evening is skippable on heavy DP weeks (6-7). Never skip DSA for LLD.
5. The 45-min timer is for focus, not a guillotine. Five minutes from cracking it? Finish. Stuck at 30 with no idea? Peek at the approach, not the code.
6. Revisit-needed problems beat new High-priority problems. A pattern you half-know is more dangerous than one you haven't started.
7. If a whole week slips, push the arc by a week — don't compress. This is 8 weeks of content, not a deadline. The plan bends so it doesn't break.

8. LLD problems: never skip the Step 5 cold re-do. Reading a solution is not practice. Rebuilding it from an empty file is.